

PROJECT CONTINUITY SUMMARY & ADVANCED BUSINESS ANALYSIS

Sample Superstore — Week 3

Prepared by: Omolade | Programme: AnalystLab Africa — Data Analytics Internship, Week 3

Part 1: Project Continuity Summary

1. The Dataset Being Used

The Sample Superstore Sales Dataset — 9,994 order-level transaction records covering 2014–2017, for a US retail company. The dataset includes order and shipping details, customer segment, product category/sub-category/name, sales, profit, discount, quantity, and geographic fields (region, state, city). This is the same dataset used in Week 2, continued into Week 3 as required.

2. The Business Problem

As a Junior Data Analyst / Business Intelligence Analyst at AnalystLab Africa Consulting, the objective is to help company leadership understand sales performance, identify profitability issues, and provide evidence-based recommendations to improve overall business health — specifically addressing why the company's 8.11% profit margin is thin, which categories or products are underperforming, and why sales appeared to decline after 2016.

3. Key Findings From Week 2

- Total Sales: \$307,800.33 | Total Profit: \$24,962.70 | Profit Margin: 8.11% | Total Orders: 1,334 | Average Sale: \$230.73
- Furniture generates strong sales but disproportionately weak profit compared to Technology and Office Supplies
- Phones and Chairs are the top-performing sub-categories by sales
- Sales peaked in 2016 and declined into 2017
- The Consumer segment drives nearly half (46.8%) of total sales
- The overall profit margin (8.11%) leaves limited room for rising costs or aggressive discounting

4. Areas Investigated Further in Week 3

- Whether the post-2016 decline was company-wide or concentrated in specific categories
- Which specific products are generating losses, not just which category
- Whether discounting is a meaningful driver of Furniture's weak profitability
- Customer-level metrics (total unique customers, average profit per order) not previously calculated
- Monthly and seasonal profit patterns, beyond the yearly view built in Week 2

Part 2: Advanced Business Analysis

This section documents the deeper analysis conducted in Week 3, organized by the areas specified in the assignment brief.

Sales Performance

Total Sales for the full period remain \$307,800.33. Breaking sales down by year and category (see dashboard Page 3) shows that overall company sales are not evenly distributed across categories or stable over time: Furniture sales fell from roughly \$30K in 2016 to \$20K in 2017, while Technology grew from about \$34K to over

\$30K over the same period, and Office Supplies remained essentially flat. High-performing periods cluster around mid-year (June–July), based on the monthly trend analysis below.

Profitability Analysis

Total Profit is \$24,962.70, an 8.11% margin. The corrected Profit Growth % measure shows profit fell 47.18% from 2016 to 2017 — a steeper decline than the sales figures alone suggest, and this is examined further below by category and product. Profit by category confirms Furniture as the weak performer: it is the only category whose profit bar visibly underperforms relative to its sales volume on the dashboard's Profit by Category chart.

Product Performance

A dedicated Loss-Making Products table was built, sorted to surface the biggest losses first. The single largest loss-maker is the Cubify CubeX 3D Printer Triple Head Print (Technology), losing \$3,839.99 on \$7,999.98 of sales — a loss rate of nearly 48% on that item. Of the seventeen largest loss-making products identified, twelve belong to the Furniture category (bookcases, tables, conference furniture, chairs), directly linking product-level losses back to the category-level Furniture problem.

Regional / Geographic Performance

Sales by Region and Sales by State visuals show reasonably even distribution across the West, East, South, and Central regions, with West and East showing marginally higher totals. No single region stands out as a major outlier the way Furniture does among categories, suggesting the profitability issue is product-driven rather than geography-driven.

Customer and Segment Analysis

A Total Customers measure was added, showing 644 unique customers across 1,334 orders — an average of roughly 2 orders per customer. Average Profit per Order was calculated at \$18.71, a useful order-level profitability benchmark. The Consumer segment continues to lead in both sales (46.8%) and order volume, consistent with Week 2 findings.

Discount Analysis

A Discount % of Sales measure shows that discounting represents just 0.07% of total sales value — a remarkably small figure. The Discount vs. Profit scatter chart, broken down by product, shows the highest-profit products cluster at low discount levels, and the largest individual losses (such as the 3D printer) are not explained by heavy discounting. This rules out discounting as a meaningful driver of Furniture's or any category's profitability problems; the cause is more likely rooted in per-unit costs such as shipping or supplier pricing.

Part 3: Summary of Advanced Analysis Findings

Analysis Area	Key Finding
Sales Performance	Furniture sales declined sharply 2016→2017; Technology grew over the same period
Profitability	Profit fell 47.18% from 2016 to 2017, driven mainly by Furniture
Product Performance	12 of the 17 biggest loss-making products are Furniture items
Regional Performance	Relatively even across regions — not a geographic issue
Customer/Segment	644 unique customers; \$18.71 average profit per order; Consumer segment leads
Discount Impact	Discounting is minimal (0.07% of sales) and does not explain the losses

Full supporting detail for these findings — including the six key business insights and five evidence-based

recommendations drawn from this analysis — is provided in the separate Business Insights and Recommendations Report submitted alongside this document.